

Ocean Guard AI

An AI-powered system for marine protection and research



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I Project Description

1 Project Overview

Ocean Guard AI is a large-scale, AI powered system designed to monitor and protect marine life in real time. It utilizes underwater drones, sonar arrays, and advanced machine learning models to detect and classify threats to marine life across the ocean. When a threat is identified, the system can alert the user enabling them to trigger safe deterrents (sound pulses or light signals) and collect valuable data for research. In addition to immediate response, Ocean Guard AI will guide a global archive of marine life interactions. This will support long-term studies on species behavior, migration as well as the impacts of climate change.

2 The Purpose of the Project

The purpose of Ocean Guard AI is to provide scientists, conservationists, and policy makers with a reliable, real-time tool for protecting vulnerable marine species and ecosystems. By combining live detection with advanced analytics, the system will not only give endangered species a chance at survival during dangerous encounters but will also generate insights that can guide conservation strategies and inform global research efforts.

2a The User Business or Background of the Project Effort

The main users of our technology are marine biologists. The “business” of this project is global marine conservation. Today, marine ecosystems face threats from pollution, climate change, overfishing, and human marine activity. Organizations such as marine research institutes, government agencies, NGOs, and fisheries are actively seeking new ways to monitor and protect ocean life. However, there is currently no real-time global monitoring system capable of detecting predator-prey interactions or providing automated responses at scale.

With Ocean Guard AI, users will be able to:

- Receive real-time alerts when vulnerable species are under threat.
- Activate safe deterrents to reduce harm without disrupting ecosystems.
- Collect long-term, reliable datasets for studies on migration, population trends, and climate change.
- Integrate findings with satellite imagery and GPS tracking for a more complete understanding of marine ecosystems.

Without a system like Ocean Guard AI, vulnerable species remain at heightened risk due to human impacts and environmental change. Developing this platform provides both immediate protective value and long-term research benefits, addressing urgent conservation needs while also laying the foundation for global scientific collaboration.

The problems faced by marine conservation are serious and urgent. Species continue to decline because current methods are too slow and limited. Without a global, real-time system, conservation teams cannot respond fast enough to threats or gather the data they need for long-term solutions. This shows the problem is significant and must be solved.

2b Goals of the Project

The goal of Ocean Guard AI is to provide a real-time, reliable, and scalable monitoring and protection system for marine ecosystems. The project aims to improve the survival chances of vulnerable marine species by giving marine biologists, conservationists, and governments actionable tools that directly enhance their work. The ultimate purpose is not just to create software, but to enable effective conservation, real-time protection, and valuable long-term ecological research.

As marine species face growing threats from human activity and climate change, current monitoring systems are too slow, localized, or reactive to make a timely difference. By setting clear goals, Ocean Guard AI avoids drifting into purely technical development and stays focused on the higher mission: protecting biodiversity and strengthening global conservation.

We want to provide marine scientists with immediate alerts when a species is under threat. We also want to allow conservation teams to intervene in real time using safe deterrents. For example, redirecting a turtle away from a polluted area using a sound deterrent.

2c Measurement

To know if our goals are met, we need clear ways to measure them. For example, one goal is that the system should detect predator-prey encounters with around 90% accuracy. We can test this by using sample data that has already been labeled by experts and then comparing our results.

Another goal is that the system should respond quickly, within 5–10 seconds, and we can measure this by checking how long it takes from detection to when an alert or deterrent is triggered.

To see if the system really helps animals, we can compare survival rates in situations before and after using Ocean Guard AI. For research, we will look at how many useful datasets are created and how many studies or reports end up using them.

Finally, to know if the system is being adopted, we can count the number of organizations like NGOs, governments, and fisheries that are using Ocean Guard AI within the first three years. It is important that these measurements are set early, because they give us a clear way to prove success and keep the project focused on its original goals.

3 The Scope of the Work

The business of this project is marine conservation, but the work of Ocean Guard AI is focused specifically on monitoring predator-prey encounters in real time and helping conservation teams respond quickly. Its role is limited to detection, alerts, deterrents, and data collection, not the entire scope of ocean conservation.

3a The Current Situation

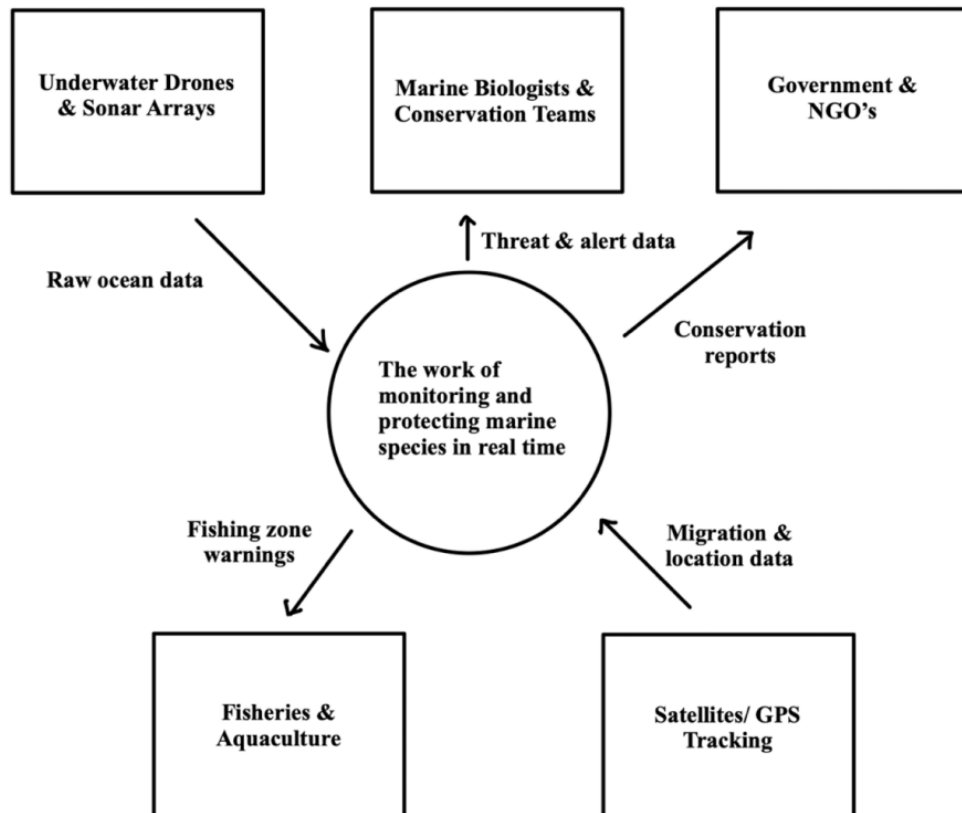
Right now, marine conservation work is mostly done through manual observation, tracking tags on animals, and occasional surveys. These methods provide useful data, but they are slow, limited to small areas, and often analyzed long after the events happen. There is no global system that can monitor threats in real time.

Because the current methods are delayed and incomplete, conservation teams often cannot act quickly enough to prevent harm to vulnerable species. By understanding these limitations, we can see why a system like Ocean Guard AI is needed to fill the gap and give faster, more reliable information.

3b The Context of the Work

The work of Ocean Guard AI is limited to monitoring predator-prey encounters, sending alerts, activating safe deterrents, and storing data for research. It does not handle larger areas of marine conservation, like setting policy or enforcing laws. To work properly, the system will need to interact with outside groups and data sources, such as marine biologists who receive alerts, government or NGO agencies that use the data, fisheries that adjust their practices, and satellite/GPS systems that provide extra environmental information.

By defining these boundaries, we make it clear what Ocean Guard AI is responsible for and what it depends on from outside systems. This helps keep the project focused and ensures the product will fit cleanly into the larger marine conservation environment.



3c Work Partitioning

Below is the list of the main business events Ocean Guard AI responds to. Each event shows the input from adjacent systems, the output, and a short summary of the business use case.

Event Name	Input	Output	Summary
Drone sends ocean data	Raw ocean data from drones/sonar	Stored data for analysis	System ingests and processes ocean data
Predator-prey encounter detected	AI classification of species behavior	Threat alerts to biologists	System detects a threat and alerts conservation teams
Deterrent activated	Operator approval or automated rule	Safe deterrent signal	System triggers a deterrent to protect the species
Satellite sends migration data	Satellite/ GPS tracking data	Updated migration maps	System refined risk zones with satellite info
Weekly report due	Scheduled task	Conservation reports to NGOs	System generates summary reports for stakeholders

These events represent the major real-world triggers that Ocean Guard AI must respond to. By breaking the work into these smaller parts, we can clearly define the system's responsibilities and later turn them into specific requirements.

3d Competing Products

There are some technologies that do parts of what Ocean Guard AI would do, but none provide the same full solution. For example, there are existing marine monitoring systems that use sonar arrays or GPS tagging to track specific species, but these are usually limited to small study areas and do not operate in real time. Drones are also used in marine research, but mainly for short missions or local observation. Some AI models are being tested for identifying species in images or videos, but they are not connected into a large network for global monitoring. None of these existing products combine drones, sonar, and AI with safe deterrents in a single, integrated system.

Looking at what already exists, we can see that Ocean Guard AI fills an important gap. Current tools are good for collecting data, but they cannot stop threats in the moment or provide a worldwide monitoring and protection system. Our project is justified because it adds real-time detection, immediate response, and large-scale integration that other products do not offer.

Existing products may still be useful as components or inspiration (for example, using current tagging systems or existing AI models as part of training). However, they are not complete solutions for marine protection, which is why Ocean Guard AI is needed.

4 The Scope of the Product

The scope of Ocean Guard AI is to serve as an intelligent monitoring and response system for marine conservation. It does not attempt to regulate entire ecosystems or replace conservation policy, but instead focuses on four key functions: detection, alerts, safe deterrence, and analytics.

For example, a marine biologist studying manatees could receive an instant alert when a shark approaches a herd, saving hours of manual video review. A conservation team monitoring sea turtles during migration could see predator hotspots and activate deterrents at the right time and place. International research institutions could share data through the system's global database, publishing joint studies that would not be possible with isolated data. Fisheries organizations could use predator activity reports to adjust fishing zones responsibly, protecting both biodiversity and their own industry's sustainability.

These scenarios illustrate the range of users and benefit that Ocean Guard AI will provide. Its scope is focused but broad enough to support conservation, research, and responsible resource management.

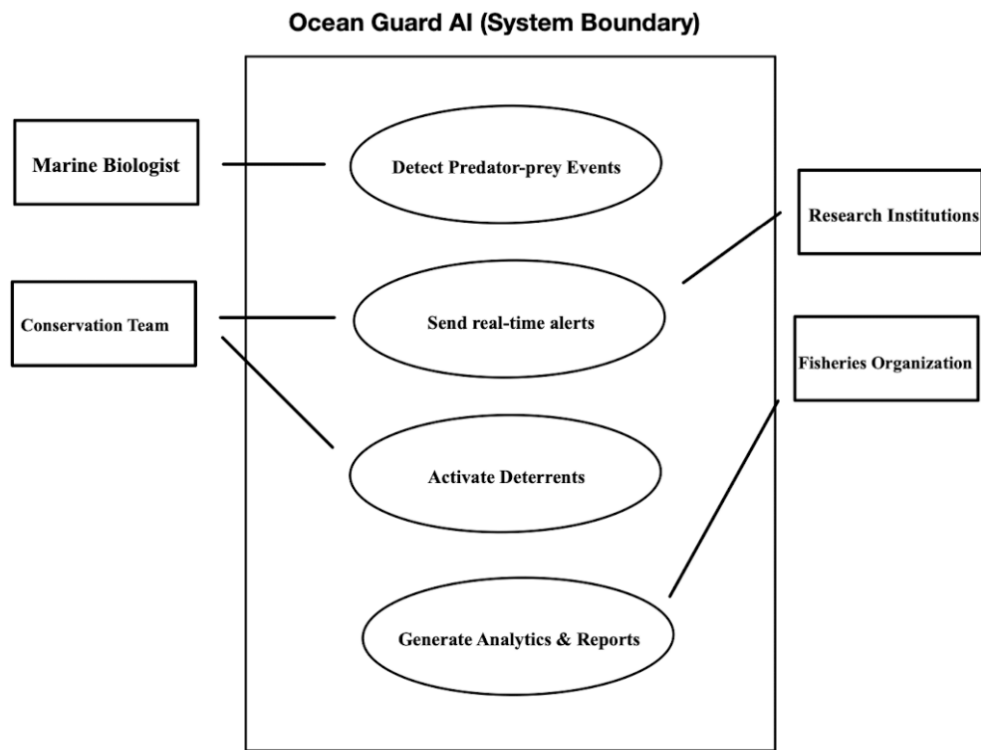
4a Scenario Diagram(s)

The scenario diagram illustrates the relationship between Ocean Guard AI and its external actors. On the outside of the system boundary are marine biologists, conservation teams, fisheries organizations, research institutions, and the marine environment itself, which provides the raw data. Inside the boundary, Ocean Guard AI processes detection, alerts, deterrents, and analytics. Arrows connect each external actor to the features they rely on, showing how the system supports real-world use. The use case diagram identifies the system's primary functions: detecting predator-prey events, sending real-time alerts, deploying safe deterrents, and providing analytics and reports. Each external actor connects only to the functions they depend on, clarifying the boundaries of the product.

4b Product Scenario List

Ocean Guard AI supports several distinct scenarios:

Marine Biologist in the Field, Marine biologist, sensor/drone network Conservation Team Monitoring Species, Conservation team, sensor/drone network, deterrent devices Global Research Collaboration, Research institutions, marine biologists Fisheries Management Application, Fisheries organizations, research institutions.



4b Product Scenario List

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Conservation Team Monitoring Species: A turtle conservation team uses sonar buoys linked to Ocean Guard AI. At night, the system detects predators near the turtles and suggests activating safe deterrents. The team reduces losses and later reviews a heatmap of high-risk areas.

Global Research Collaboration: Researchers in Canada and Japan upload data to the system's shared database. The AI highlights similar predator strategies across oceans, leading to a joint publication and advancing global science.

Fisheries Management Application: A fisheries group in Chile uses Ocean Guard AI's weekly reports to track predator activity. When predators cluster around anchovy schools, they adjust fishing zones to protect stocks and preserve sustainability.

5 Stakeholders

The Client

The client for Ocean Guard AI would likely be a marine conservation organization or government agency such as WWF, NOAA, or UNEP. These groups provide funding and strategic direction, motivated by protecting biodiversity and advancing conservation technology.

The Customer

Customers include conservation groups, government agencies, fisheries organizations, and research institutions. They would purchase Ocean Guard AI to gain more effective tools for real-time monitoring, long-term reporting, and resource management.

Hands-On Users

Hands-on users include marine biologists, conservation teams, fisheries managers, and research scientists. Each group has different expertise and technology skills but shares the need for accurate detection, usable dashboards, and actionable data. Marine biologists and researchers contribute scientific expertise, while conservation teams rely on usability in the field. Fisheries managers focus on sustainability through analytics.

Maintenance Users

Maintenance users include system administrators, developers, and field technicians who install, update, and service the system. They require robust documentation, modular design, and durable hardware.

Other Stakeholders

Other stakeholders include sponsors, testers, business analysts, technology experts, legal advisors, and usability experts. International associations such as UNEP or IUCN also have a role in ensuring that the system fits into global conservation efforts. Recognizing these groups ensures that requirements are not missed and that the system gains broad support.

User Participation

Users must be actively involved throughout development. Marine biologists and conservation teams will test AI accuracy and usability in the field, while fisheries and research institutions will shape reporting standards. Maintenance users will define requirements for sustainability. Without this participation, requirements risk being incomplete or misaligned.

User Priorities

The most important users are marine biologists and conservation teams, as they depend on the system for daily conservation work. Research institutions and fisheries organizations are secondary users whose input is valuable but not critical to core success. The public may benefit indirectly, but their needs are not prioritized in design.

6 Mandated Constraints

However, building this Ocean Guard AI system, we will face certain obstacles that can halt or slow down the progression of the development. We have to account for multiple different constraints including technology functionality, environmental challenges, device connections, budgeting of the project, and the limited time of development. We have taken the constraints that will come with developing the project into account and thought of some ideas to resolve these conflicts.

6a Solution Constraints

First, we need to think about the data collecting devices that we plan on using for this project. We have underwater drones to collect data about marine life which also plays a role in correctly identifying and classifying certain species in the ocean. They will observe and gather information about marine life, like their homes, diets, predators, and lifestyle.

Now these drones will contain AI models and algorithms to manage the data collection and management in the databases of the system. The constraints about this are the accuracy of these AI models. It needs to be made sure that these AI models are collecting and identifying information correctly regarding threats to their environment and way of life. Misidentifying certain threats under the system could lead to false positives and ineffective safe deterrents due to misinterpretation of the AI models. These AI models need to be investigated consistently for accurate information gathering.

The networks of the oceans are unstable. This means that our underwater drones may need to operate offline, instead of online Wi-Fi-dependencies. This is mainly because there are more challenges of communication and limitation of wireless signals underwater. We need to establish other forms of network functionality offline to prevent loss of drone control, delayed responses, malfunctioning GPS navigations, and low radio connection. Operating under Wi-Fi networks, our drones are highly at risk of losing connection to our control system, mapping navigations, and even losing potential data acquired by those specific drones.

Importantly to note as well, the storage capacity of these drones needs to be reasonable for these huge amounts of data collection. We don't want to miss out on our data collections because the drones ran out of storage space. Under the ocean depths, the transferring of data to the system might require higher speeds of communication and connection. The computing of these underwater drones needs to be able to perform and function properly under the unstable network connection in these environments. We could have trouble retrieving real-time data that is up to date and cause delays in sending real threat alerts when these are not accounted for.

6b Implementation Environment of the Current System

The marine conditions could create problems for our underwater drones. The further down these underwater drones go into the ocean abyss, the more pressure they would withhold. With a large amount of underwater pressure, the hardware of the underwater drones has to be shielded from pressure to prevent hardware damage to our data collectors. They must be built with pressure-resistant and waterproof capabilities.

Internal components under harsh pressures of the ocean can implode due to high amounts of pressure.

The temperature of the ocean depth holds potential threats to the drone's battery life. Though these drones are designed for underwater environments, extreme water temperatures can still affect the electronics, battery, and sensors of these machines. This is kind of like how aerial drones are affected by air temperature regarding battery efficiency and electronic functionality. We would require our underwater drones to have a reasonable battery life that will allow them to function underwater for long periods of time. We want these drones to operate more than 24 hours underwater as extreme cold temperatures can drain battery life.

Salinity and corrosive factors have potential effects on underwater drones that need to be resolved. The salinity (Salt Concentration) and other corrosive factors boost up the degradation of materials embedded into our drones. Degradation of materials can cause electrical failures, hardware issues, and low structural capacity against high pressure. High amounts of salinity or salt in the water increase the amount of electrical conductivity of the device. With high amounts of electrical conductivity, the water conducts more electrochemical corrosion turning the water environment into one aggressive electrolyte that facilitates even more corrosive factors.

6c Partner or Collaborative Applications

The collaboration and partnering of NGOs and research institutions are vital for this project's progression. Partnering with research institutions and NGOs, we can acquire expert validated data regarding underwater wildlife and expand coverage globally. It can provide technological development and resources to help develop the needs of the system. Because we see our stakeholders to include expert researchers on marine biology and conservation, Ocean Guard AI is something that these research institutions can use.

When implementing Ocean Guard AI, we need to be mindful of global policy on environmental principles. It should be assured that Ocean Guard AI is compliant to underlying regulations of collective actions, rules, and agreements to address environmental challenges. Governmental agencies like the US Environmental Protection Agency and the National Oceanic Atmospheric Administration play a key role in enforcing laws and regulations when it comes to developing things that might interfere. Ocean Guard AI needs to account for any form of unethical use of the environment, especially if it violates environmental regulations.

It would also be helpful if we could get Ocean Guard AI access to satellite networks. Because underwater drones and sonar arrays only cover a tiny, limited area, the use of satellites can help expand the system's coverage area. Satellites allow GPS tracking of marine animals and migration patterns in distances beyond our devices' reach. Satellites work as relay channels, where they acquire data from one destination then transport it to another destination over a vast distance. It allows our underwater drones to transmit collected data faster to labs and the system.

In relation to satellites, they collect information on marine life analysis including sea surface temperatures, food availability, and pollution levels in ecosystems. As many conservationists have already used satellites to track marine animals, integration of Ocean Guard AI is a good step to expanding it.

6d Off-the-Shelf Software

Due to the large-scale data storage, the monitoring system and drones will rely on, we plan to use PostgreSQL, Docker, and TensorFlow. However, constraints to this might come with compatibility and scalability issues. Not all software, including the ones listed, work smoothly with underwater drone devices and sonar arrays especially if it is done in a single data management system/. They might require more than just one software to function properly and assume additional custom adapters and APIs. PostgreSQL and Docker have limitations to their scalability where handling more massive data in a large area might go beyond what they can handle when collecting real time data.

As mentioned before, the network connections in underwater environments are unstable. This creates problems with our planned software as they might assume a kind of internet Wi-Fi or cloud access function. As we have talked in the previous paragraphs that Ocean Guard AI might need to operate in an offline setting, we need to find a way that the off-the-shelf software we are using can operate in the respective setting. This is something that software platforms may not fully support and might be challenging to make it work without the internet.

Importantly, sensitive data especially from research and governmental institutions needs to be safeguarded with the planned software that we are using. The Ocean Guard AI system needs to be protected from cyber-attacks and hackers. Because data in satellite and underwater drones is transmitted wirelessly, they can be intercepted and compromised. Privacy is another thing that needs to be accounted for, especially policy regulations in data protection.

6e Anticipated Workplace Environment

For Ocean Guard AI, it will work in multiple work environment collaborations. The workplace environments under Ocean Guard AI are a mix of labs, ships, research facilities, and offices for human workers. Then we also have our device-controlled workers like drones, sonar arrays, and satellites combining the efforts of underwater, surface level, and space. Now this would make the scale of the project very large. The splitting of the work environments into many can create more time in development because software and hardware's used would have to be tested in both controlled conditions of the labs and uncontrolled conditions at sea.

In return we would also have to gather a large number of professionals in different professions to develop the perfect marine life monitoring system. This would take the combined work of multiple different teams all collaborating on one goal. In our case, the following teams being considered would be Research, Programming, Management, Data Analytics, Engineering, Network, Environmental Policy, Interface Designers, and Ship Crews.

The research team consists of marine biologists, ecologists, and conservationists actively researching and studying marine life. The programming team will deal with developing the backend of the systems as well as the AI models used to classify marine life. The management team is responsible for resource management and budgeting of the project. Data analytics will consist of data scientists and analysts analyzing and tracking the datasets being collected. The engineering team will be a group of robotics and hardware engineers building the devices and keeping them properly functioning.

The network team will oversee establishing network solutions so that satellites, sonar arrays, and underwater drones can communicate with one another effectively even in unstable situations. The environmental policy team will be responsible for keeping the system in check in terms of regulations and compliance with laws. Interface designers will deal with building the interfaces for Ocean Guard AI to build dashboards and easy to interpret visualizations. Lastly, we have our crew members operating in ships often deploying underwater drones and using sonar arrays to gather information.

The sheer number of different professions Ocean Guard AI needs would require more emphasis on interdisciplinary teamwork of professions, instead of a single software project. Safety protocols would also need to be initiated in cases of bad weather situations in the sea for ship crew workers. Safety regulations can also limit our testing grounds for deploying sonar arrays and drones, which is why the environmental policy team is important. Because our workers would be so spread out, we would also need to account for different time zones and the effects that they can have in decision making.

6f Schedule Constraints

As you can see, the development of Ocean Guard AI is a long-term development project adding some time constraint to it with expected deadlines. The process of systems programming, technological readiness, environmental testing, and data collecting would consume a lot in a span of limited time to build the most we can of Ocean Guard AI. Because these systems cannot be fully built and ready to go all at once, it is best to break up the development into multiple different stage phases. These stage phases must be completed in chronological order, but with deadlines to get the most we can for the whole semester.

To add as well, because the separate locations of multiple teams working collaboratively in labs, ships, offices, and research facilities would be so spread out, the scheduling might take a bit more effort considering that the time zones will not be the same for every worker for Ocean Guard AI. The cross-team collaboration of developers, marine biologists, and even the partnering institutions would have to synchronize scheduling plans and testing periods. In consequence, this could probably slow down communication and approval by a lot.

Importantly, the testing periods and deployment of these data collecting devices and sonar arrays would have to be really well planned out. First these devices needed to be tested before being actually used and faced potential delays in their development due to environmental challenges, functionality issues, inaccuracy, a limited access to testing vessels, and environmental regulations. After testing, the scaling to additional regions, refining AI accuracy, and securing data infrastructure would be carefully thought out before actually implementing them.

6g Budget Constraints

With Ocean Guard AI, a large-scale project, would require an immense amount of budget due to developing underwater drones, AI models, satellites usage, global data structures, and multiple teams of different professions. Because of the size of the work environment and resources, we estimate that a fully functioning Ocean Guard AI system would cost millions of dollars to create, around \$5 million to \$15 million to create. This is one of the major critical constraints to our project.

The manufacturing of the underwater drones and sonar arrays would be really expensive. Implementing hardware that is resistant to pressure and implosion plus the harsh temperatures of the depths, we would need high-quality materials and precise engineering. The development of the AI models would require a lot of money to train and maintain so that they can process data in unstable underwater connections, fast computation, and large data storages.

Then there comes the cost of developing our labs, research facilities, and ships. The need for field testing, renting ships, acquiring the necessary equipment and logistics of deploying drones would add a couple hundred thousand more to the budget. Because we are considering partnering with research institutions, organizations, and services, we would need to account for the fees of collaboration and accessing their resources.

7 Naming Conventions and Definitions

7a Definitions of Key Terms

Entity	Attribute	Definition
Species	SpeciesID	Unique identifier for each species
	Name	Common name of the species
	ScientificName	Scientific classification name
	Predators	Species that prey on this species
	Prey	Species consumed by this species
	Diet	Dietary type (carnivore, herbivore, omnivore)
	RiskLevel	Endangerment or protection status
Encounters	EncounterID	Unique identifier for encountering events
	SpeciesInvolved	List of species participating in the encounter
	LocationOfEncounter	Geographic location where the encounter occurred
	TimeOfEncounter	Timestamp of the encounter
	TypeOfEncounter	Timestamp of the encounter
	Description	Additional details or notes

Alerts	AlertID	Unique identifier for each alert
	EncounterID	Reference to related encounter
	PriorityLevel	Urgency level of the alert
	SpeciesThreatened	Species at risk in the alert
	SpeciesThreat	Species or factor causing the threat
Drones	DeviceID	Species or factor causing the threat
	DeviceType	Type of drone
	DeviceLocation	Geographic coordinates or assigned location
	DataCollected	Information gathered (video, sonar, environmental)
	DeviceStatus	Operational status
Monitors	DeviceID	Unique identifier for each sonar/ sensor device
	NetworkCoverageArea	Geographic area monitored by the device
	DeviceLocation	Placement of the device
	DeviceSensitivity	Sensitivity level of detector
	DeviceStatus	Operational status
	DetectedEncounters	List of encounters detected
	DetectedMovements	Movement patterns detected
	Description	Additional notes
Users	UserID	Unique identifier for each user
	UserProfession	Role or profession of the user
	PhoneNumber	Contact phone number
	Email	Contact email address

	StartDate	Date when the user registered in the system
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7b UML and Other Notation Used in This Document

The system uses UML notation to describe components, actors, and data flows. Arrows represent data transfer, actors represent users (e.g., biologists, NGOs), and the system boundary outlines Ocean Guard AI's scope. Consistent notation ensures all diagrams are interpreted the same way by different stakeholders. If notation is only used once (e.g., in a specific diagram), it should be explained directly in that section. This document follows the OMG UML 2.0 standard, with exceptions noted where applicable.

7c Data Dictionary for Any Included Models

The data dictionary explains the main information Ocean Guard AI uses. For species, we keep track of IDs, names, diets, and risk levels. Encounters store details like which species were involved, where and when it happened, and what type of interaction occurred. Alerts connect to encounters and show the urgency, the species at risk, and the cause of the threat. Drones and monitors are described by their IDs, location, status, and the data they collect. Finally, users are recorded with their ID, role, and contact information.

The dictionary makes the system's information clear, so everyone knows exactly what each piece of data means. This avoids confusion and keeps the project consistent. The data dictionary helps both analysts and developers. Analysts define the important terms, while developers can later add the technical details when building the system.

8 Relevant Facts and Assumptions

8a Facts

Important background information for the Ocean Guard AI project:

- No real-time global monitoring system exists today. The world lacks operational systems which enable real-time observation of marine life.
- Marine biologists, conservationists, and governments are actively looking for technologies that protect marine life.
- The world needs technological solutions which will help protect marine ecosystems and preserve biodiversity.
- Climate change and human impacts have increased biodiversity loss over the years. The rapid decrease in marine biodiversity requires immediate action for Ocean Guard AI development because of its critical importance.

8b Assumptions

The following assumptions are made to guide the system design:

- Under harsh ocean conditions, underwater drones and sonar arrays will operate properly and effectively. It is assumed that the system's hardware can withstand high pressure, salinity, and other challenging marine environments.
- AI models demonstrate the ability to extract precise and dependable information for species classification and identification tasks. The system will generate precise information which researchers and conservation teams can use for their work.
- Global regulations and ethical compliance regarding environmental protection and endangered species will be satisfied. The project follows all international laws and regulations and ethical standards that protect marine life.

II Requirements

9 Product Use Cases

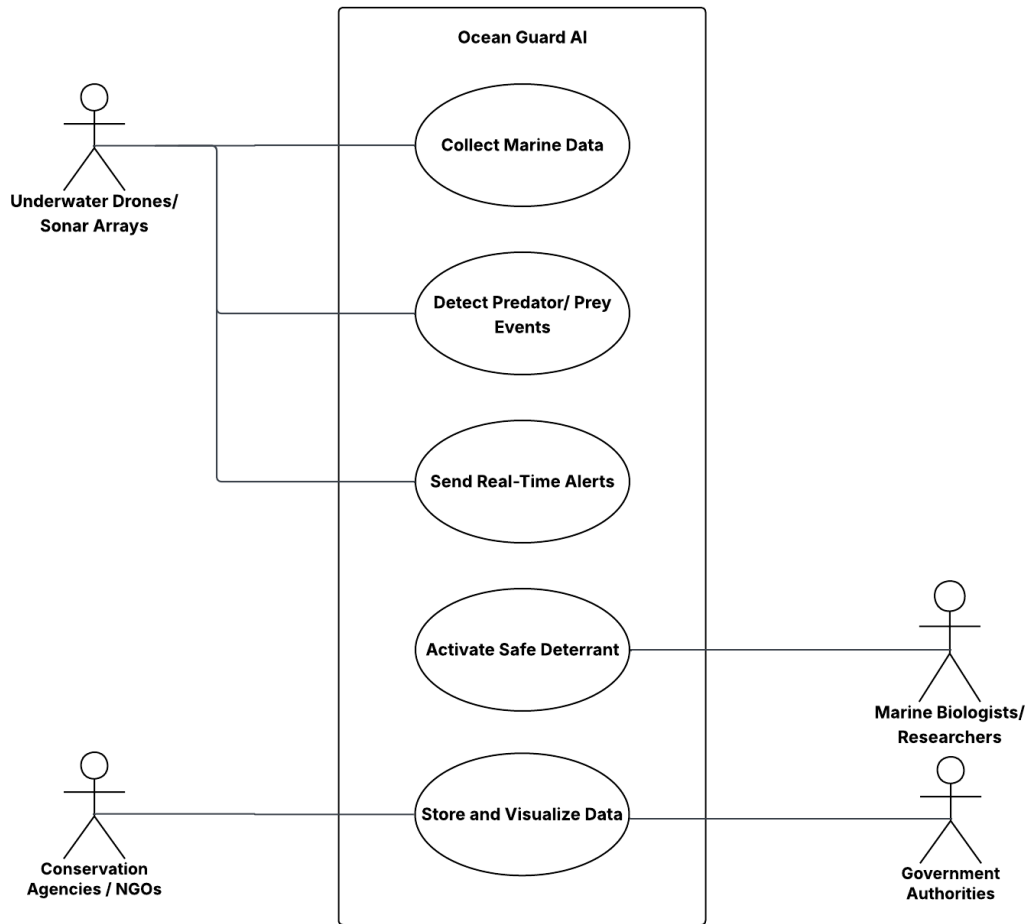
Ocean Guard AI is an AI powered monitoring system that helps protect ocean life through real-time detection, analysis, and response. It utilizes underwater drones, sonar arrays, and AI models to recognize threats, alert researchers, and collect valuable ecological data.

9a Use Case Diagrams

The Ocean Guard AI Use Case Diagram shows how people and external devices (actors) interact with the system. It defines what is inside the system (the software that runs Ocean Guard AI) and what is outside (the users, drones, and agencies that use it).

List of Primary Actors:

- **Marine Biologists / Researchers** – Monitor marine environments and activate safe deterrents.
- **Underwater Drones / Sonar Arrays** – Gather data on ocean activity and send alerts.
- **Conservation Agencies / NGOs** – Access reports, monitor species activity, and evaluate environmental protection efforts.
- **Government Authorities** – Oversee compliance with national and international environmental policies.



Created with Lucidchart by Meleena Torres – OceanGuard AI Use Case Diagram

9b Product Use Case List

Use Case ID: UC-01

Use Case Name: Collect Marine Data

Primary Actors: Underwater Drones / Sonar Arrays

Description: Ocean Guard AI gathers continuous sonar, visual, and environmental data from underwater sensors and drones to monitor marine activity and habitat conditions.

Use Case ID: UC-02

Use Case Name: Detect Predator/ Prey Events

Primary Actors: Underwater Drones / Sonar Arrays

Description: The system uses AI-powered analysis to identify predator-prey interactions by recognizing movement patterns, sounds, and proximity between marine species.

Use Case ID: UC-03

Use Case Name: Send Real-Time Alerts

Primary Actors: Underwater Drones / Sonar Arrays

Description: When potential threats or rare encounters are detected, Ocean Guard AI automatically sends real-time notifications to researchers and connected monitoring stations for immediate awareness.

Use Case ID: UC-04

Use Case Name: Activate Safe Deterrents

Primary Actors: Marine Biologists / Researchers

Description: Upon receiving an alert, users can remotely activate non-harmful deterrents such as sound pulses or light flashes to protect vulnerable animals from predators or human-related dangers.

Use Case ID: UC-05

Use Case Name: Store and Visualize Data

Primary Actors: Conservation Agencies / NGOs, Government Authorities

Description: Ocean Guard AI compiles long-term data on marine encounters and presents it through analytical dashboards, helping organizations study population trends, migration routes, and climate impacts.

9c Individual Product Use Cases

Use Case ID: UC-01

Use Case Name: Collect Marine Data

Pre-conditions: System and underwater drones are active and connected.

Post-conditions: Collected data is stored in the Ocean Guard database.

Initiated by: Underwater Drones / Sonar Arrays

Triggering Event: Drone begins a scheduled data collection cycle.

Additional Actors: Ocean Guard AI (Data Processor)

Sequence of Events:

1. Drone activates sensors and starts recording ocean movement, sound and temperature. The system receives this information in real time.
2. The drone sends sonar and camera data to the Ocean Guard AI server. The system checks and cleans the data to remove errors or noise.
3. Drone completes the scan cycle. System stores all validated data to be analyzed later.

Alternatives:

- If the signal drops, the drone connects to a nearby node.
- A user can manually change how deep the drone collects data.

Exceptions:

- The drone cannot send data because of a hardware problem.
- The server is temporarily down and cannot save data.

Use Case ID: 02

Use Case Name: Detect Predator/ Prey Events

Pre-conditions: Underwater drones and sonar arrays are actively collecting data and transmitting it to the Ocean Guard AI system.

Post-conditions: The system finds and records possible predator-prey encounters in the Ocean Guard AI database.

Initiated by: Underwater Drones / Sonar Arrays

Triggering Event: New live data shows movement or sound patterns that may indicate animal interactions.

Additional Actors: Ocean Guard AI Detection Module (AI Processor)

Sequence of Events:

1. Drones send sonar and camera data to the Ocean Guard AI system. The system cleans and organizes the data (removes noise, adds timestamps).
2. Drones keep scanning the area. The system uses AI to recognize and identify nearby marine species.
3. The AI notices unusual movement or distance between animals that could mean a predator-prey event. The system labels the type of event and gives it a confidence score.
4. If the confidence score is high enough, the system saves the event details and sends them to the alert system.

Alternatives:

- AI flags the event for manual review if confidence level < threshold.
- System aggregates similar detections for pattern analysis before confirming an event.

Exceptions:

- Data corrupted or lost during transmission.
- AI model fails to identify species due to poor visibility or incomplete data.
- Detection module offline or return processing error.

Use Case ID: 03**Use Case Name:** Send Real-Time Alerts

Pre-conditions: The AI system has detected a predator-prey event with a high enough confidence score. The system is connected to the network and can reach registered users.

Post-conditions: Alerts are successfully sent to marine biologists and other authorized users. The alert details are saved in the system.

Initiated by: Ocean Guard AI System

Triggering Event: A confirmed detection of a predator-prey event requires immediate attention.

Additional Actors: Marine Biologists / Researchers

Sequence of Events:

1. The system prepares an alert with event details like location, time, and confidence level.
2. The system checks who should receive the alerts. It formats and sends messages through a dashboard notification.
3. Users receive the alert and review the event information. The system records when users open or confirm the alert.

Alternatives:

- If several events happen at once, the system sends alerts starting with the most serious ones.
- If the connection is slow, the system retries sending the alert using another server.

Exceptions:

- No internet connection, so the alert cannot be sent.
- The user's account is inactive or lacks access to permissions.
- The event is later found to be a false detection.

Use Case ID: 04**Use Case Name:** Activate Safe Deterrents

Pre-conditions: The system has confirmed a predator-prey event and is connected to the deterrent devices.

Post-conditions: The deterrent is activated safely, and the event is logged in the system database.

Initiated by: Marine Biologists / Researchers

Triggering Event: A user receives an alert and chooses to activate a deterrent, or the system automatically triggers it when a high-risk event occurs.

Additional Actors: Ocean Guard AI (Deterrent Control Module)

Sequence of Events:

1. A confirmed predator-prey event is detected. The system provides the user with the option to activate a deterrent.
2. The user approves activation, or the system triggers it automatically. The deterrent sends out sound pulses or light signals underwater.
3. The event is recorded as "deterrent activated." The system saves all related data, including time, duration, and response type.

Alternatives:

- The user may choose not to activate a deterrent and only observe the event.
- The system can delay activation to verify the detection first.

Exceptions:

- The deterrent fails to activate due to a hardware or connection error.
- The system triggers the deterrent too late or repeats activation unnecessarily.

Use Case ID: 05

Use Case Name: Store and Visualize Data

Pre-conditions: The Ocean Guard AI system is active and receives valid data from drones or event logs.

Post-conditions: All collected and processed data is stored and displayed through visual dashboards for analysis.

Initiated by: Conservation Agencies / NGOs, Government Authorities

Triggering Event: The system finishes processing a data collection or detection cycle and prepares results for viewing.

Additional Actors: Marine Biologists / Researchers

Sequence of Events:

1. The system receives verified data from previous operations. It stores the information in the main Ocean Guard AI database.
2. The system organizes the data by location, time, and event type. It updates visual dashboards and graphs for users to view trends and summaries.
3. Authorized users log in to review reports and event details. The system displays the latest data clearly and accurately.

Alternatives:

- Users can export visualized data into files or reports.
- The system can automatically share summaries with partner organizations.

Exceptions:

- Data fails to be saved due to a storage issue.
- Visualization tools load slowly or fail to update correctly.
- User access permissions prevent viewing certain datasets.

10 Functional Requirements

ID# - F-01 - Data Collection from Drones

Description: The system must continuously collect sonar, camera, and environmental data from all connected underwater drones and sonar arrays, then store it in the Ocean Guard AI database.

Rationale: The system needs to collect data constantly so it can detect activity and understand what's happening in the ocean.

Fit Criterion: The requirement is met when the system successfully receives and stores data from connected drones without major errors or interruptions.

Acceptance Tests: AT-01 (Drone Connection Test), AT-02 (Data Storage Test)

ID# - F-02 - AI-Based Event Detection

Description: The system uses artificial intelligence to analyze incoming data and detect predator-prey interactions between marine animals.

Rationale: Automated detection helps researchers identify events quickly without having to manually watch large amounts of data.

Fit Criterion: The requirement is met when the AI correctly recognizes sample predator-prey encounters and records them accurately.

Acceptance Tests: AT-03 (AI Detection Test)

ID# - F-03 - Alert Generation and Notification

Description: When the AI detects an important event, the system sends an alert to researchers through email, dashboard messages, or mobile notifications.

Rationale: Real-time alerts help users act quickly and stay informed about what's happening in monitored areas.

Fit Criterion: The requirement is met when alerts are automatically sent after an event is confirmed and can be received by authorized users.

Acceptance Tests: AT-04 (Alert Delivery Test)

ID# - F-04 - Activate Safe Deterrents

Description: The system allows researchers or the AI to trigger safe deterrents, such as light or sound pulses, to protect vulnerable species.

Rationale: Deterrents give the system a safe and fast way to reduce risk for marine animals during dangerous encounters.

Fit Criterion: The requirement is met when a deterrent can be activated manually or automatically, and the action is logged correctly in the system.

Acceptance Tests: AT-05 (Deterrent Activation Test)

ID# - F-05 - Data Storage and Visualization

Description: The system stores collected data and shows it through charts and maps on a dashboard that users can access.

Rationale: Visualizing information helps users analyze long-term trends and make better conservation decisions.

Fit Criterion: The requirement is met when stored data appears correctly on the dashboard and matches the information in the database.

Acceptance Tests: AT-06 (Dashboard Display Test)

11 Data Requirements

ID# - D-01 - Marine Observation Data

Description: This includes sonar readings, camera images, and ocean details such as depth, temperature, and location collected by underwater drones.

Rationale: The system needs this data to understand ocean conditions and find marine activity.

Fit Criterion: This requirement is met when drones send full, clear data that is saved correctly and can be viewed later.

Acceptance Tests: AT-01 (Drone Connection Test), AT-02 (Data Storage Test)

ID# - D-02 - Species Identification Data

Description: Information about different marine species found by the system, such as names, size, type, and behaviors.

Rationale: The system needs to recognize and label animals correctly to tell what kind of interaction is happening.

Fit Criterion: This requirement is met when the AI can name or classify animals accurately during testing.

Acceptance Tests: AT-03 (AI Detection Test)

ID# - D-03 - Event Detection Records

Description: A record of detected predator-prey encounters, including the animals involved, time, location, and related sonar or camera data.

Rationale: Saving this information helps researchers study how animals behave and check that the AI is working right.

Fit Criterion: This requirement is met when events are automatically recorded and easy to find later.

Acceptance Tests: AT-03 (AI Detection Test), AT-04 (Alert Delivery Test)

ID# - D-04 - Alert and Deterrent Logs

Description: Details about alerts and deterrent actions, such as who got the alert, when it was sent, and what response happened.

Rationale: Keeping these logs helps track how the system reacts and makes sure it works safely.

Fit Criterion: This requirement is met when every alert or deterrent action is saved and matches the related event data.

Acceptance Tests: AT-04 (Alert Delivery Test), AT-05 (Deterrent Activation Test)

ID# - D-05 - Dashboard and Visualization Data

Description: Summarized data shown on maps and charts, like animal movement, event counts, or ocean trends.

Rationale: This helps users see important patterns and understand what is happening in the ocean.

Fit Criterion: This requirement is met when dashboard data shows correctly and matches what's stored in the system.

Acceptance Tests: AT-06 (Dashboard Display Test)

12 Performance Requirements

12a Speed and Latency Requirements

ID# - P-01 - Real-Time Detection Speed

Description:

Ocean Guard AI must detect and process marine events quickly while data is being received from underwater drones.

Rationale: Fast detection lets researchers respond to threats in real time and prevents delays in alerts.

Fit Criterion: The system should analyze and report a detected event within a few seconds after receiving the data.

Acceptance Tests: AT-03 (AI Detection Test), AT-04 (Alert Delivery Test)

12b Precision or Accuracy Requirements

ID# - P-02 - Accurate Event Detection

Description: The AI must correctly identify marine species and classify predator-prey events with high reliability.

Rationale: Accurate identification avoids false alerts and ensures valid data for research.

Fit Criterion: During testing, the AI should correctly identify species and event types in most sample cases.

Acceptance Tests: AT-03 (AI Detection Test)

12c Capacity Requirements

ID#: P-03 - Concurrent Data Handling

Description: The system should be able to handle data from several underwater drones by sending information at the same time.

Rationale: Multiple drones may operate in different areas, so the system must process all their data without slowing down or crashing.

Fit Criterion: The requirement is met when the system can process multiple live data streams smoothly during testing.

Acceptance Tests: AT-01 (Drone Connection Test), AT-02 (Data Storage Test)

13 Dependability Requirements

13a Reliability Requirements

ID#: DEP-01 - Reliability Requirements

Description: The system must stay running and not crash during drone operations or data uploads.

Rationale: Continuous operation is needed, so no ocean data is lost.

Fit Criterion: Systems can run for long monitoring sessions (8+ hours) without shutting down or producing major errors.

Acceptance Tests: AT-01 (Drone Connection Test), AT-02 (Data Storage Test)

13b Availability Requirements

ID#: DEP-02 - 24/7 System Access

Description: Ocean Guard AI should always be available for use by drones and agencies.

Rationale: Marine monitoring is constant, so the system must stay online.

Fit Criterion: System uptime should be over 99%, including automatic restart after brief downtime.

Acceptance Tests: AT-01 (Drone Connection Test)

13c Robustness or Fault-Tolerance Requirements

ID#: DEP-03 - Offline Operation Mode

Description: If network or server connections fail, drones should keep collecting data and upload it later.

Rationale: Offline data collection prevents data loss when communication fails.

Fit Criterion: System stores local data safely during outages and uploads it when the link returns.

Acceptance Tests: AT-02 (Data Storage Test)

13d Safety-Critical Requirements

ID#: DEP-04 - Safe Deterrent Activation

Description: When a risk event happens, deterrents must operate without harming marine life or equipment.

Rationale: To protect animals and follow environmental safety laws.

Fit Criterion: Deterrent tests show all actions stay within safe decibel and light-level limits.

Acceptance Tests: AT-05 (Deterrent Activation Test)

14 Maintainability and Supportability Requirements

14a Maintenance Requirements

ID#: MAIN-01 - Easy System Updated

Description: Developers should be able to update or patch the system without interrupting operations.

Rationale: This keeps Ocean Guard AI secure and up to date.

Fit Criterion: Updates installed successfully with minimal downtime.

Acceptance Tests: AT-07 (System Update Test)

14b Supportability Requirements

ID#: MAIN-02 - User Help and Monitoring Support

Description: Provide help guides and quick tech support for researchers and agencies using the dashboard.

Rationale: Support ensures users can handle problems without disrupting monitoring.

Fit Criterion: Help tools and contact info are available directly in the interface.

Acceptance Tests: AT-08 (User Support Test)

14c Adaptability Requirements

ID#: MAIN-03 – Multi-Platform Compatibility

Description: System should work on desktops, laptops, and tablets.

Rationale: Different users operate from labs, ships, or offices.

Fit Criterion: Interface and features function the same across devices.

Acceptance Tests: AT-09 (Cross-Platform Test)

14d Scalability or Extensibility Requirements

ID#: MAIN-04 – Expand to New Regions

Description: The System must be able to add new monitoring zones or drone units easily.

Rationale: Supports growth of ocean monitoring operations over time.

Fit Criterion: New drones or locations can be added without major reconfiguration.

Acceptance Tests: AT-10 (System Expansion Test)

14e Longevity Requirements

ID#: MAIN-05 - Long-Term Operation

Description: Ocean Guard AI should stay reliable for several years with regular maintenance.

Rationale: Ensures the system continues to serve research goals and justifies development cost.

Fit Criterion: The system remains functional and updatable for at least 5 years.

Acceptance Tests: AT-11 (Longevity Test)

15 Security Requirements

15a Access Requirements

ID#: SEC-01 – Authorized User Access

Description: Only approved users (researchers, agencies, or admins) can view or change system data.

Rationale: Limits unauthorized access and keeps data safe.

Fit Criterion: Login permissions correctly restrict access to each role.

Acceptance Tests: AT-12 (User Access Test)

15b Integrity Requirements

ID#: SEC-02 – Authorized User Access

Description: Only approved users (researchers, agencies, or admins) can view or change system data.

Rationale: Limits unauthorized access and keeps data safe.

Fit Criterion: Login permissions correctly restrict access to each role.

Acceptance Tests: AT-12 (User Access Test)

15c Privacy Requirements

ID#: SEC-03 - User Privacy and Confidential Data

Description: Private information from users for agencies must remain confidential.

Rationale: Follows data protection laws and builds user trust.

Fit Criterion: System hides personal or location data unless authorized

Acceptance Tests: AT - 14 (Privacy Protection Test)

15d Audit Requirements

ID#: SEC-04 - Activity Log and Tracking

Description: All user and system actions should be logged in for accountability and traceability.

Rationale: Audit logs help track issues and prevent misuse

Fit Criterion: Each login and data change is stored with user ID and timestamp.

Acceptance Tests: AT - 15 (Audit Log Test)

15e Immunity Requirements

ID#: SEC-05 - System Protection from Malware

Description: The system must detect and block any virus or unauthorized program.

Rationale: Prevents attacks that could stop data collection or harm devices.

Fit Criterion: System passes routine malware and penetration testing

Acceptance Tests: AT - 16 (Security Defense Test)

16 Usability and Humanity Requirements

16a Ease of Use Requirements

ID#: U1 – Easy-to-Use Interface

Description: The system shall provide a simple and clear interface, so biologists and field workers can quickly view alerts, maps, and animal information without confusion.

Rationale: Users may be in field environments and need to react quickly without learning complex menus.

Fit Criterion: At least 80% of test users shall complete basic tasks in under 2 minutes after a brief walkthrough.

Acceptance Tests: AT-17 – Interface Navigation Test

ID#-U2-Low User Errors

Description: The system shall help users avoid mistakes such as ignoring urgent alerts or clicking the wrong action.

Rationale: Incorrect decisions could put animals at risk.

Fit Criterion: User errors remain below 2% during pilot testing/

Acceptance Tests: AT-18 – Error Prevention Test

ID# - U3 - Clear Feedback on Actions

Description: The system shall provide visual confirmation for user actions.

Rationale: Users need confidence that their actions worked.

Fit Criterion: Feedback within 5 seconds.

Acceptance Tests: AT-19 – Action Feedback Test

ID# - U4 - Easy to Learn and Remember

Description: Users should learn the system quickly without long training.

Rationale: Some users are volunteers or seasonal workers.

Fit Criterion: 75% complete tasks after a 30-minute introduction.

Acceptance Tests: AT-20 – Usability Learning Test

16b Personalization and Internationalization Requirements

ID# - ID-P1 - Language and Regional Options

Description: System shall support English at launch and allow future expansion to other languages at least 10 languages.

Rationale: International researchers may need the local language

Fit Criterion: Language setting available in UI settings.

Acceptance Tests: AT-21 – Language Settings Test

ID# - ID-P2 - User Display Preferences

Description: Users can adjust map layers and alert display format.

Rationale: Different users prioritize different information.

Fit Criterion: User settings are saved and applied to login.

Acceptance Tests: AT-22 – Display Preference Test

16c Learning Requirements

ID# - ID-L1 - Quick Learning Curve

Description: Users shall be able to operate the main functions after a short walkthrough.

Rationale: Field use requires fast onboarding.

Fit Criterion: 75% perform core tasks after a 30–45-minute intro.

Acceptance Tests: AT-23 – Training Simulation Test

ID# - ID-L2 - Built-in Guided Help

Description: Basic tips or icons available to guide new users.

Rationale: Helps seasonal/volunteer users learn quickly.

Fit Criterion: Help icons present in major UI areas.

Acceptance Tests: AT-24 – Guided Help Icon Test

16d Understandability and Politeness Requirements

ID# - Ud1 - Intuitive Marine Terminology

Description: Terminology match real marine biology practices.

Rationale: Users should understand the system without technical confusion.

Fit Criterion: 80% users agree terms are clear.

Acceptance Tests: AT-25 – Terminology Clarity Survey

ID# - Ud2 - Simple Alerts and Messages

Description: Alerts use plain language (ex, “Shark near turtle zone.”).

Rationale: Field workers need a fast understanding.

Fit Criterion: Messages readable under 5 seconds.

Acceptance Tests: AT-26 – Alert Readability Test

16e Accessibility Requirements

ID# - Ac1 - Colorblind-Friendly Design

Description: System shall avoid red/green-only status colors.

Rationale: 20% of males are red-green colorblind.

Fit Criterion: Accessibility audit passes.

Acceptance Tests: AT-27 – Color Accessibility Test

ID# - Ac2 – Large Text Option

Description: Users can enable larger text

Rationale: Field screens & older users may need bigger text.

Fit Criterion: Large text toggle available.

Acceptance Tests: AT-28 – Text Size Adjustment Test

16f User Documentation Requirements

ID# - Doc1 – Online User Guide

Description: Web help page with basic steps and FAQs.

Rationale: Users can review instructions anytime.

Fit Criterion: Guide linked to app.

Acceptance Tests: AT-29 – Online Help Access Test

ID# - Doc2 – Quick Start Sheet

Description: 1-page PDF for new users.

Rationale: Fast onboarding for field teams.

Fit Criterion: One-page guide available.

Acceptance Tests: AT-30 – Quick Guide Access Test

16g Training Requirements

ID# - Tr1 – Basic Training Session

Description: Provide optional 1-hour intro training.

Rationale: Ensures all user types feel comfortable.

Fit Criterion: Training outlines exist.

Acceptance Tests: AT-31 – Training Completion Test

17 Look and Feel Requirements

17a Appearance Requirements

ID# - Ap1 – Clean and Professional Interface

Description: Interface shall use ocean-themed colors and simple fonts.

Rationale: Clarity needed for field use.

Fit Criterion: User review scores $\geq 4/5$.

Acceptance Tests: AT-32 – Interface Review Test

ID# - Ap2 – Consistent Icon & Label Style

Description: Icons & labels follow the standard guide.

Rationale: Prevent confusion.

Fit Criterion: 100% icons consistent.

Acceptance Tests: AT-33 – Visual Consistency Test

17b Style Requirements

ID# - St1 – Scientific, Trustworthy Tone

Description: Professional, research-oriented UI.

Rationale: Builds trust in data accuracy.

Fit Criterion: 70% of users rate it trustworthy.

Acceptance Tests: AT-34 – Trustworthiness Feedback Test

ID# - St2 – Minimal Distraction Design

Description: Avoid flashy effects.

Rationale: Field use requires focus.

Fit Criterion: Limited pop-ups.

Acceptance Tests: AT-35 – Focus Retention Test

18 Operational and Environmental Requirements

18a Expected Physical Environment

ID# - Env1 – Outdoor Readability

Description: UI must work in sunlight and noisy areas.

Rationale: Field research happens outside.

Fit Criterion: 85% users confirm visibility.

Acceptance Tests: AT-36 – Sunlight Visibility Test

ID# - Env2 – Offline Mode

Description: Store recent data for offline use.

Rationale: Remote ocean areas may lack a signal.

Fit Criterion: Stores 10 minutes of data.

Acceptance Tests: AT-02 – Data Storage Test

18b Requirements for Interfacing with Adjacent Systems

ID# - Int1 – Drone & Sensor Integration

Description: Support data input from drones/sensors.

Rationale: Need for real-time wildlife monitoring.

Fit Criterion: Connects to at least 1 drone and buoy.

Acceptance Tests: AT-01 – Drone Connection Test

ID# - Int2 – GIS Export Support

Description: Export to CSV/JSON.

Rationale: Research data sharing.

Fit Criterion: Works with ArcGIS and Sheets.

Acceptance Tests: Data Export Test

18c Productization Requirements

ID# - Prod – Web-Based Access

Description: Web-Based Access

Rationale: Faster onboarding.

Fit Criterion: Access for under 60 seconds.

Acceptance Tests: AT-38 – Login Speed Test

18d Release Requirements

ID# - Re11 – Scheduled Updates

Description: Updates twice per year after release.

Rationale: AI and field features improve

Fit Criterion: Documented update schedule

Acceptance Tests: AT-07 – System Update Test

19 Cultural and Political Requirements

19a Cultural Requirements

ID# - C-01- Cultural Neutrality

Description: Ocean Guard AI will use neutral symbols, colors, and terminology in its interface and documentation. The system will avoid language or visuals that could offend or confuse users from different cultural groups.

Rationale: The platform serves global research teams, so cultural neutrality ensures equal usability for all users and supports international collaboration.

Fit Criterion: At least five partner organizations from different regions confirm that the interface and documentation are culturally acceptable.

Acceptance Tests: AT-06 (Usability Test), AT-21 (Language Settings Test)

ID# - C-02- Multilingual Accessibility

Description: The system will support at least five languages—English, French, Spanish, Japanese, and Korean.

Rationale: Multilingual support helps global users interpret results accurately.

Fit Criterion: Native speakers review translations to confirm clarity and accuracy.

Acceptance Tests: AT-21 (Language Settings Test), AT-07 (Multilingual Interface Validation)

19b Political Requirements

ID# - P-01- Compliance with Environmental Policies

Description: The system must comply with international ocean protection and anti-pollution regulations.

Rationale: Ensure global credibility and alignment with conservation laws.

Fit Criterion: Review reports confirm that system operations meet environmental standards.

Acceptance Tests: AT-04 (Environmental Compliance Test)

ID# - P-02- Political Neutrality

Description: Ocean Guard AI must present ocean data objectively, avoiding support for any country's political stance.

Rationale: Neutrality maintains trust and prevents conflict among participating nations.

Fit Criterion: Independent reviewers confirm that reports and dashboards show no political bias.

Acceptance Tests: AT-04 (Environmental Compliance Test)

ID# - P-03- Respect for National Data Boundaries

Description: The system will not share or transmit a nation's marine data without its consent.

Rationale: Prevents international disputes and respects national data ownership.

Fit Criterion: Regional access controls prevent unauthorized data transfers.

Acceptance Tests: AT-05 (Data Security Test)

20 Legal Requirements

20a Compliance Requirements

ID# - L-01- User and Location Data Protection

Description: The system will protect personal and location data, following global privacy standards.

Rationale: Builds trust and ensures compliance with international data protection laws.

Fit Criterion: Validation confirms that private data stays internal until users grant access.

Acceptance Tests: AT-05 (Data Security Test)

ID# - L-02- Environmental Law Compliance

Description: The system will follow environmental regulations that safeguard marine life and water quality.

Rationale: Supports lawful conservation efforts and sustainable research.

Fit Criterion: Verified documentation shows compliance with both local and global ocean laws.

Acceptance Tests: AT-04 (Environmental Compliance Test)

20b Standards Requirements

ID# - S-01- Secure Development Standards

Description: The system will follow structured, secure coding and testing procedures throughout development.

Rationale: Promotes reliability and minimizes risks during updates.

Fit Criterion: Review confirms adherence to secure-development best practices.

Acceptance Tests: AT-05 (Data Security Test)

ID# - S-02- Consistent Data Formats

Description: Ocean Guard AI will use standardized ocean-data formats for compatibility with research tools

Rationale: Enables smooth data exchange and reduces conversion errors.

Fit Criterion: Test exports open correctly in tools such as ArcGIS and spreadsheets.

Acceptance Tests: AT-37 (Data Export Test)

21 Requirements Acceptance Tests

21a Requirements – Test Correspondence Summary

Table 1a – Functional & Data Requirements – Acceptance Test Correspondence

Acceptance Test ID	F-01	F-02	F-03	F-04	F-05	D-01	D-02	D-03	D-04	D-05
AT-01 Drone Connection Test	X					X				
AT-02 Data Storage Test	X					X				
AT-03 AI Detection Test		X					X	X		
AT-04 Alert Delivery Test			X					X	X	
AT-05 Deterrent Activation Test				X					X	
AT-06 Dashboard Display Test					X					X

Figure 1. Functional and Data Requirements Shows how the system collects, saves, and displays drone data correctly.

Table 1b – Performance & Dependability Requirements – Acceptance Test Correspondence

Acceptance Test ID	P-01	P-02	P-03	DEP-01	DEP-02	DEP-03	DEP-04
AT-01			X	X	X		
AT-02			X	X		X	
AT-03	X	X					
AT-04	X						
AT-05							X

Figure 2. *Performance and Dependability Requirements Shows how the system keeps good speed, accuracy, and reliability.*

Table 1c – Maintainability & Security Requirements – Acceptance Test Correspondence

Acceptance Test ID	MAIN- 01	MAIN- 02	MAIN- 03	MAIN- 04	MAIN- 05	SEC- 01	SEC- 02	SEC- 03	SEC- 04	SEC- 05
AT-07 System Update Test	X									
AT-08 User Support Test		X								
AT-09 Cross- Platform Test			X							
AT-10 System Expansion Test				X						
AT-11 Longevity Test					X					
AT-12 User Access Test						X				
AT-13 Data Integrity Test							X			
AT-14 Privacy Protection Test								X		
AT-15 Audit Log Test									X	
AT-16 Security Defense Test										X

Figure 3. *Maintainability and Security Requirements Shows how updates, user help, and data protection work in the system.*

Table 1d – Usability & Learning Requirements – Acceptance Test Correspondence

Acceptance Test ID	U1	U2	U3	U4	ID-P1	ID-P2	ID-L1	ID-L2
AT-17 Interface Navigation Test	X							
AT-18 Error Prevention Test		X						
AT-19 Action Feedback Test			X					
AT-20 Usability Learning Test				X				
AT-21 Language Settings Test					X			
AT-22 Display Preference Test						X		
AT-23 Training Simulation Test							X	
AT-24 Guided Help Icon Test								X

Figure 4. *Usability and Learning Requirements Shows how the system is easy for users to learn and use.*

**Table 1e – Accessibility, Documentation, Training, and Look & Feel Requirements –
Acceptance Test Correspondence**

Acceptance Test ID	Ud1	Ud2	Ac1	Ac2	Doc1	Doc2	Tr1	Ap1	Ap2	St1	St2	Env1
AT-25 Terminology Clarity Survey	X											
AT-26 Alert Readability Test		X										
AT-27 Color Accessibility Test			X									
AT-28 Text Size Adjustment Test				X								
AT-29 Online Help Access Test					X							
AT-30 Quick Guide Access Test						X						
AT-31 Training Completion Test							X					
AT-32 Interface Review Test								X				
AT-33 Visual Consistency Test									X			
AT-34 Trustworthiness Feedback Test										X		
AT-35 Focus Retention Test											X	
AT-36 Sunlight Visibility Test												X

Figure 5. *Accessibility, Documentation, Training, and Look & Feel Requirements Shows how the system stays accessible, clear, and visually consistent.*

**Table 1f – Operational, Environmental, Cultural, Political, Legal, and Standards Requirements
– Acceptance Test Correspondence**

Acceptance Test ID	Env2	Int1	Int2	Prod	Re11	C-01	C-02	P-01	P-02	P-03
AT-02 Data Storage Test	X									
AT-01 Drone Connection Test		X								
AT-37 Data Export Test			X							
AT-38 Login Speed Test				X						
AT-07 System Update Test					X					
AT-06 Usability Test						X				
AT-21 Language Settings Test							X			
AT-04 Environmental Compliance Test								X	X	
AT-05 Data Security Test										X

Figure 6. *Operational, Cultural, Political, Legal, and Standards Requirements Shows how the system works across environments, cultures, and follows rules and standards.*

21b Acceptance Test Descriptions

AT-01 – Drone Connection Test

Checks that the system can connect to underwater drones and begin receiving data.

AT-02 – Data Storage Test

Verifies that collected sonar and camera data are stored correctly and can be retrieved.

AT-03 – AI Detection Test

Confirms that the AI can recognize predator-prey events when provided with sample data.

AT-04 – Alert Delivery Test

Tests that alerts are sent automatically to authorized users once an event is detected.

AT-05 – Deterrent Activation Test

Ensures that safe deterrents can be triggered manually or automatically when needed.

AT-06 – Dashboard Display Test

Checks that stored data appears correctly in charts, maps, and tables on the dashboard.

AT-07 – System Update Test

Checks that developers can install software updates or patches without interrupting ongoing monitoring.

AT-08 – User Support Test

Verifies that users can access help materials or contact support through the system interface.

AT-09 – Cross-Platform Test

Ensures that the system runs properly on different devices (desktop, tablet, or laptop) with consistent performance.

AT-10 – System Expansion Test

Tests that new drones or monitoring zones can be added without changing the core setup or causing system errors.

AT-11 – Longevity Test

Simulates long-term operation to confirm that the system stays stable and maintainable over time.

AT-12 – User Access Test

Checks that only authorized users (e.g., researchers, admins) can log in and view or change specific data.

AT-13 – Data Integrity Test

Verifies that data cannot be altered or deleted without permission and that all edits are tracked.

AT-14 – Privacy Protection Test

Ensures that private user or organization information is stored securely and hidden from unauthorized users.

AT-15 – Audit Log Test

Checks that all system and user actions (logins, edits, alerts) are recorded with timestamps for traceability.

AT-16 – Security Defense Test

Simulates malware or intrusion attempts to confirm that the system detects and blocks unauthorized access.

AT-17 – Interface Navigation Test

Checks that users can open maps, view data, and trigger alerts without confusion.

AT-18 – Error Prevention Test

Verifies that the system prevents common mistakes (e.g., wrong clicks or missed alerts).

AT-19 – Action Feedback Test

Ensures the system provides clear confirmation messages or visual feedback after each user action.

AT-20 – Usability Learning Test

Tests that new users can complete basic actions after a short walkthrough.

AT-21 – Language Settings Test

Confirms that users can change languages and see translations correctly.

AT-22 – Display Preference Test

Checks that personal map or alert display preferences save and reload correctly.

AT-23 – Training Simulation Test

Verifies that users can learn the main functions using demo or tutorial mode.

AT-24 – Guided Help Icon Test

Ensures that help icons and tooltips appear in all key areas of the UI.

AT-25 – Terminology Clarity Survey

Collects user feedback to confirm marine biology terms are understood easily.

AT-26 – Alert Readability Test

Checks that alerts are short, clear, and readable in under five seconds.

AT-27 – Color Accessibility Test

Ensures that the interface passes a color-blindness accessibility audit.

AT-28 – Text Size Adjustment Test

Confirms that users can toggle larger text and it applies to all screens.

AT-29 – Online Help Access Test

Checks that the online user guide opens from the system's help menu.

AT-30 – Quick Guide Access Test

Ensures that the one-page PDF quick start guide is downloadable and readable.

AT-31 – Training Completion Test

Verifies that all participants finish the basic 1-hour training with positive feedback.

AT-32 – Interface Review Test

Collects user ratings to confirm the UI looks clean and professional.

AT-33 – Visual Consistency Test

Checks that all icons, buttons, and labels match the design standard.

AT-34 – Trustworthiness Feedback Test

Surveys users to confirm that the interface feels credible and research oriented.

AT-35 – Focus Retention Test

Ensures that no excessive pop-ups or distractions interrupt normal use.

AT-36 – Sunlight Visibility Test

Tests that screens are readable in bright outdoor light.

AT-37 – Data Export Test

Verifies that exported files open correctly in ArcGIS and spreadsheet tools.

AT-38 – Login Speed Test

Measures time from login to dashboard load and confirms it's under 60 seconds.